

Model card for the Capstone Project

1. Model overview

Model name: Hybrid Bayesian Optimisation Framework

Version: v1.0

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This model is a hybrid Bayesian Optimisation framework designed to optimise the performance of eight unknown, black-box objective functions under a limited evaluation budget. The system combines Gaussian Process surrogate modelling with dynamic acquisition functions and neural network weighted candidate selection. Gaussian Processes with Radial Basis Function (RBF) and Matérn kernels are used to model the objective function response surface, while a neural network with a straightforward architecture estimates the probability of candidate points belonging to high-performing regions in the search space. These probabilities are incorporated into the acquisition function calculation to strengthen search efficiency.

Core optimiser process: Gaussian Process (RBF/Matérn) → Neural-network-derived probability weighting → Acquisition Function (UCB/EI) → Next candidate selection.

2. Intended use

Primary task:

Optimisation of eight benchmark black-box objective functions.

Target users:

Researchers, student, or practitioners working on optimisation problems where evaluations are computationally expensive or limited.

Recommended use cases:

- Hyperparameter optimisation
- Simulation-based optimisation
- Financial/trading parameter tuning
- General black-box optimisation challenges

Not recommended for:

- Real-time control systems requiring immediate decisions
- Problems where the objective function is known analytically
- Safety-critical systems without extensive validation
- Extremely high-dimensional optimisation problems without additional dimensionality reduction techniques.

The model is intended to improve sampling efficiency by identifying promising regions of the candidate search space by balancing exploration and exploitation.

3. Training data

Data sources:

Data is generated through evaluations of candidate solutions submitted to the black-box optimisation environment.

Size of data set:

The dataset grows incrementally throughout optimisation. Each observation consists of an input parameter vector and corresponding objective value. With 13 total submissions, the size of each of the datasets will grow by 13 data points over the course of the project.

Table 1: Dataset size breakdown by objective function

Objective Function	Number of input features (dimensionality)	Number of initial data points provided	Final size of the dataset
1	2	10	23
2	2	10	23
3	3	15	28
4	4	30	43
5	4	20	33
6	5	20	33
7	6	30	43
8	8	40	53

Languages or modalities:

Numerical input parameters and numerical objective values.

Preprocessing steps:

- Output values were scaled in some instances to improve surrogate modelling
- Labelling of high-performing regions for neural network training

The neural network is trained on previously evaluated candidate points and corresponding objective values to estimate the likelihood of future candidates belonging to promising regions.

4. Evaluation metrics

Metrics used:

For optimisation performance:

- Best observed output value per iteration
- Improvement per step
- Step distance between consecutive query points
- Uncertainty at sampled points

For the neural network:

- Training loss
- Training accuracy

Performance results:

Performance is measured by comparing objective values achieved across iterations and by analysing convergence across different acquisition function and surrogate model configurations.

Fairness or bias checks:

Traditional biases do not exist in this project since data is purely numerical and does not represent any human-derived information. However, in terms of search space bias, queries have been focused in local areas due to:

- 1) The tendency of Bayesian Optimisation algorithms to focus on high-performing local regions.
- 2) The limited number of queries, making it unfeasible to explore the entire search space.

Known limitations:

- Gaussian Process performance may degrade as dimensionality increases.
- Acquisition functions may over-exploit promising regions.
- Neural network predictions are based on the quality, quantity, and diversity of previously sampled points.
- Objective functions are assumed to be smooth, which is a vital property for Gaussian Process surrogate modelling.

5. Ethical considerations

Potential biases or risks:

- Overfitting to observed regions in the search space
- Convergence in local optima whilst missing global optimum
- Reduced exploration due to inaccurate neural network predictions

Mitigation strategies:

- Use of uncertainty-aware acquisition functions
- Comparison of multiple acquisition functions (UCB vs EI)
- Continued exploration of uncertain regions

Privacy concerns:

No personal, sensitive, or human-generated data is used. The optimisation process operates solely on numerical parameter vectors and objective function outputs.

6. Model life cycle

Date of last update: 04/06/2026

Version control or repository: GitHub repository with full version history and experiment tracking.

Monitoring plan:

- Track optimisation performance across iterations
- Track model changes and adaptations throughout the project and include in documentation to maintain transparency and repeatability